MongoDB Practice Task (Products & Orders Collections)

Question 1

Find all products that belong to the Electronics category and have a price greater than ₹5000.

```
db.products.find({
  category: "Electronics",
  price: { $gt: 5000 }
})
```

Question 2

Display each order with the Product Name, Price, and Quantity.

```
db.orders.aggregate([
  {
    $lookup: {
      from: "products",
      localField: "product_id",
      foreignField: "_id",
      as: "product_details"
    }
  },
  {
    unwind: "product_details"
```

```
  },
  {
    $project: {
      _id: 0,
      productName: "$product_details.name",
      price: "$product_details.price",
      quantity: 1
    }
  }
])
```

Question 3

Find all products whose stock is between 20 and 80 (inclusive).

```
db.products.find({
  stock: {
    $gte: 20,
    $lte: 80
  }
})
```

Question 4

Calculate the average price of products in each category. Display only:

- **Category**
- **Average Price**

Sort the result by average price in descending order.

```
db.products.aggregate([
  {
    $group: {
      _id: "$category",
      averagePrice: { $avg: "price" }
    }
  },
  {
    $project: {
      _id: 0,
      category: "$_id",
      averagePrice: 1
    }
  },
  {
    $sort: {
      averagePrice: -1
    }
  }
])
```

Question 5

Find all products that are not in the Electronics category.

db.products.find({

```
    category: { $ne: "Electronics" }
})
```

Question 6

Display each order with:

- **Product Name**
- **Category**
- **Price**
- **Quantity**

Sort the results by product price in descending order.

```
db.orders.aggregate([
  {
    $lookup: {
      from: "products",
      localField: "product_id",
      foreignField: "_id",
      as: "productDetails"
    }
  },
  {
    unwind: "productDetails"
  },
  {
    $project: {
```

```
        _id: 0,
        productName: "$productDetails.name",
        category: "productDetails.category", price: "productDetails.price",
        quantity: 1
      }
    },
    {
      $sort: {
        price: -1
      }
    }
])
```

Question 7

Find all products whose price is less than ₹3000 and stock is greater than 20.

```
db.products.find({
  price: { $lt: 3000 },
  stock: { $gt: 20 }
})
```

Question 8

Calculate the total inventory value for each product.

Formula:

Total Value = Price × Stock

```
db.products.aggregate([
```

```
  {
    $project: {
      _id: 0,
      productName: "$name",
      category: 1,
      totalInventoryValue: {
        multiply: ["price", "$stock"]
      }
    }
  },
  {
    $sort: {
      totalInventoryValue: -1
    }
  }
])
```

Display:

- **Product Name**
- **Category**
- **Total Inventory Value**

Sort by Total Inventory Value in descending order.

Question 9

Find all products whose category is either Electronics or Clothing, and whose price is greater than ₹1000.

```
db.products.find({
  category: {
    $in: ["Electronics", "Clothing"]
```

```
  },
  price: { $gt: 1000 }
})
```

Question 10

Calculate the total amount for every order.

Formula:

Total Amount = Product Price × Quantity

Display:

- **Product Name**
- **Quantity**
- **Price**
- **Total Amount**

Sort by Total Amount in descending order.

```
db.orders.aggregate([
  {
    $lookup: {
      from: "products",
      localField: "product_id",
      foreignField: "_id",
      as: "product_details"
    }
  },
```

```
    {
        unwind: "product_details"
    },
    {
        $project: {
            _id: 0,
            productName: "$product_details.name",
            quantity: 1,
            price: "$product_details.price",
            totalAmount: {
                multiply: ["product_details.price",
                    "$quantity"
                ]
            }
        }
    },
    {
        $sort: {
            totalAmount: -1
        }
    }
])
```